

## Lab 2: Empirical Risk and Simple Linear Regression

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 9th, 2026

Name: \_\_\_\_\_

username: \_\_\_\_\_

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all required activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

### Recap: The Modeling Recipe

In [Chapter 1.3](#), we introduced the three-step modeling recipe for finding optimal model parameters, which ultimately helps us make the best possible predictions.

#### 1. Choose a model.

$$\underbrace{h(x_i) = w}_{\text{constant model}} \quad \underbrace{h(x_i) = w_0 + w_1 x_i}_{\text{simple linear regression model}}$$

#### 2. Choose a loss function.

$$\underbrace{L_{\text{sq}}(y_i, h(x_i)) = (y_i - h(x_i))^2}_{\text{squared loss}} \quad \underbrace{L_{\text{abs}}(y_i, h(x_i)) = |y_i - h(x_i)|}_{\text{absolute loss}}$$

#### 3. Minimize *average loss* (also called *empirical risk*) to find optimal model parameters.

- Constant model, squared loss:  $R_{\text{sq}}(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w)^2 \implies w^* = \bar{y}$
- Constant model, absolute loss:  $R_{\text{abs}}(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w| \implies w^* = \text{Median}(y_1, y_2, \dots, y_n)$
- Simple linear regression model, squared loss:

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2 \implies w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

### Activity 1: Relative Squared Loss

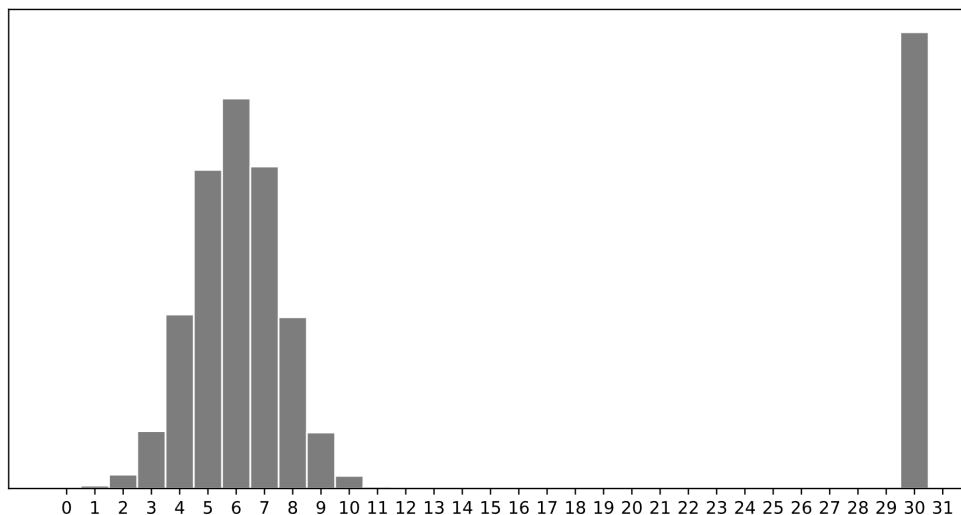
Suppose we'd like to find the optimal parameter,  $w^*$ , for the constant model  $h(x_i) = w$ . To do so, we use the following loss function, called the **relative squared loss**:

$$L_{\text{rsq}}(y_i, h(x_i)) = \frac{(y_i - h(x_i))^2}{y_i}$$

What value of  $w$  minimizes the average loss (i.e. empirical risk) when using the relative squared loss function – that is, what is  $w^*$ ? Your answer should only be in terms of the variables  $n, y_1, y_2, \dots, y_n$ , and any constants.

## Activity 2: Rapid Fire

Consider a dataset of  $n$  **integers**,  $y_1, y_2, \dots, y_n$ , whose histogram is given below:



a) Which of the following is closest to the constant prediction  $w^*$  that minimizes:

$$\frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = w \\ 1 & y_i \neq w \end{cases}$$

- ☐ 1   ☐ 5   ☐ 6   ☐ 7   ☐ 11   ☐ 15   ☐ 30

b) Which of the following is closest to the constant prediction  $w^*$  that minimizes:

$$\frac{1}{n} \sum_{i=1}^n |y_i - w|$$

- ☐ 1   ☐ 5   ☐ 6   ☐ 7   ☐ 11   ☐ 15   ☐ 30

c) Which of the following is closest to the constant prediction  $w^*$  that minimizes:

$$\frac{1}{n} \sum_{i=1}^n (y_i - w)^2$$

- ☐ 1   ☐ 5   ☐ 6   ☐ 7   ☐ 11   ☐ 15   ☐ 30

d) Which of the following is closest to the constant prediction  $w^*$  that minimizes:

$$\lim_{p \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n |y_i - w|^p$$

*Hint: Think about the effect of outliers.*

- ☐ 1   ☐ 5   ☐ 6   ☐ 7   ☐ 11   ☐ 15   ☐ 30

### Activity 3: Slope of Mean Absolute Error

Consider a dataset of 8 points,  $y_1, y_2, \dots, y_8$  that are in sorted order, i.e.  $y_1 < y_2 < \dots < y_8$ .

Recall that mean absolute error,  $R_{\text{abs}}(w)$ , for the constant model  $h(x_i) = w$  is defined as:

$$R_{\text{abs}}(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|$$

This is a piecewise linear function that changes slope at each data point. The slope of  $R_{\text{abs}}(w)$  at any  $w$  that is not a data point is:

$$\frac{d}{dw} R_{\text{abs}}(w) = \frac{\text{number left of } w - \text{number right of } w}{n}$$

Suppose that  $y_4 = 10$ ,  $y_5 = 14$ ,  $y_6 = 22$ , and  $R_{\text{abs}}(11) = 9$ . What is  $R_{\text{abs}}(22)$ ?

*Hint: You don't have all 8 of the  $y$ -values, so you can't find  $R_{\text{abs}}(22)$  just by plugging in numbers into the formula for  $R_{\text{abs}}(w)$ . Instead, think about how to use the slope formula.*

## Activity 4: Programming

Complete the tasks in the `lab02.ipynb` notebook.

There are two ways to access the supplemental Jupyter Notebook:

- **Option 1 (preferred):** Set up a Jupyter Notebook environment locally, use `git` to clone our [course repository](#), and open `labs/lab02/lab02.ipynb`. For instructions on how to do this, see the [Environment Setup](#) page of the course website.
- **Option 2:** Click [here](#) to open `lab02.ipynb` on DataHub. Before doing so, read the instructions on the [Environment Setup](#) page on how to use the DataHub.

To receive credit for Activity 4, you'll need to show your lab TA that all test cases have passed.

## Activity 5: What Do You Mean?

Suppose we want to fit a simple linear model (using squared loss) that predicts the number of ingredients in a product given its price. We're given that:

- The average cost of a product in our dataset is \$40, i.e.  $\bar{x} = 40$
- The average number of ingredients in a product in our dataset is 15, i.e.  $\bar{y} = 15$

The intercept and slope of the regression line are  $w_0^* = 11$  and  $w_1^* = \frac{1}{10}$ , respectively.

- a) Suppose Victors' Veil (a skincare product) costs \$40 and has 11 ingredients. What is the squared loss of our model's predicted number of ingredients for Victors' Veil?

- b) Is it possible to answer part a) above **just** by knowing  $\bar{x}$  and  $\bar{y}$ , i.e. **without** knowing the values of  $w_0^*$  and  $w_1^*$ ? Once you select an answer, explain it to your peers.

*Hint: If you're stuck, look through the headings of the sections of [Chapter 2.3](#).*

☐ Yes, it's possible    ☐ No, it's not possible

## Activity 6: The Meaning of Mean Squared Error

Suppose we'd like to predict the number of minutes a delivery will take,  $y$ , as a function of distance,  $x$ . To do so, we look to our dataset of  $n$  deliveries,  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ , and fit two simple linear models:

- $F(x_i) = a_0 + a_1 x_i$ , where:

$$a_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

Here,  $\bar{x}$  and  $\bar{y}$  are the respective means of the  $x$ -values and  $y$ -values.

- $G(x_i) = b_0 + b_1 x_i$ , where  $b_0$  and  $b_1$  are chosen such that  $G(x_i) = b_0 + b_1 x_i$  minimizes **mean absolute error** on the dataset. Assume that no other line minimizes mean absolute error on the dataset, i.e. that the values of  $b_0$  and  $b_1$  are unique. Assume also that  $F$  and  $G$  are distinct lines.

a) Fill in the

$$\sum_{i=1}^n (y_i - F(x_i))^2 \quad \text{} \quad \sum_{i=1}^n (y_i - G(x_i))^2$$

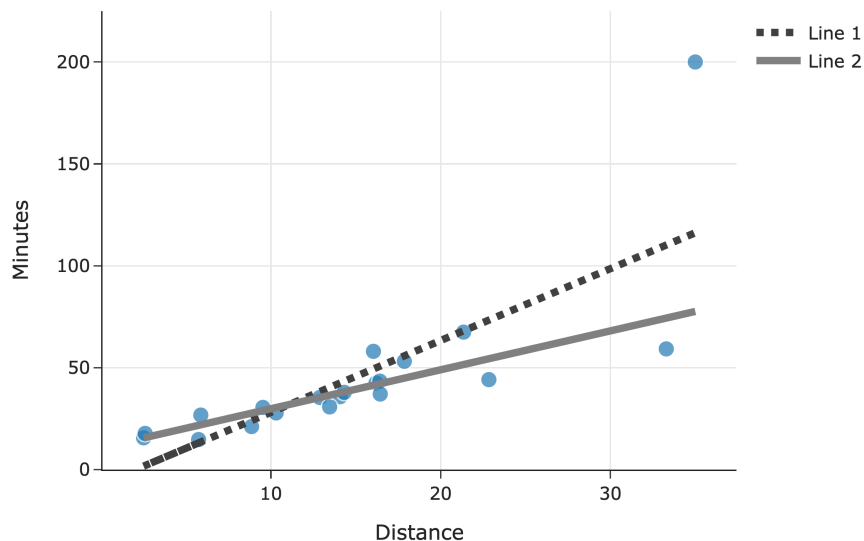
☐ >   ☐ ≥   ☐ =   ☐ ≤   ☐ <   ☐ Impossible to tell

b) Fill in the

$$\left( \sum_{i=1}^n |y_i - F(x_i)| \right)^2 \quad \text{} \quad \left( \sum_{i=1}^n |y_i - G(x_i)| \right)^2$$

☐ >   ☐ ≥   ☐ =   ☐ ≤   ☐ <   ☐ Impossible to tell

c) Below, we've drawn the lines for both  $F$  and  $G$  along with a scatter plot for the original  $n$  deliveries:



Which line corresponds to  $F$ ?

☐ Line 1

☐ Line 2

The following are extra practice. Don't feel pressured to answer all of these problems in lab, but make sure to attempt them at some point.

### Activity 7: Relative Squared Loss, Continued

Recall the formula for **relative squared loss** from Activity 1:

$$L_{\text{rsq}}(y_i, h(x_i)) = \frac{(y_i - h(x_i))^2}{y_i}$$

- a) Let  $C(y_1, y_2, \dots, y_n)$  be your minimizer  $w^*$  from Activity 1. That is, for a particular dataset  $y_1, y_2, \dots, y_n$ ,  $C(y_1, y_2, \dots, y_n)$  is the value of  $w$  that minimizes empirical risk for relative squared loss on that dataset.

What is the value of  $\lim_{y_4 \rightarrow \infty} C(1, 3, 5, y_4)$  in terms of  $C(1, 3, 5)$ ? Your answer should involve the function  $C$  and/or one or more constants.

*Hint: To notice the pattern, evaluate  $C(1, 3, 5, 100)$ ,  $C(1, 3, 5, 10000)$ , and  $C(1, 3, 5, 1000000)$ .*

- b) What is the value of  $\lim_{y_4 \rightarrow 0} C(1, 3, 5, y_4)$ ? Again, your answer should involve the function  $C$  and/or one or more constants.

- c) Based on the results of the previous two parts, when is the prediction  $C(y_1, y_2, \dots, y_n)$  robust to outliers? When is it not robust to outliers?

### Activity 8: A Refresher

Consider a dataset of  $y_1, y_2, \dots, y_n$ , all of which are **positive**. We want to fit a constant model,  $h(x_i) = w$ , to the data.

Let  $w_p^*$  be the optimal constant prediction that minimizes average degree- $p$  loss,  $R_p(w)$ , defined below:

$$R_p(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^p$$

For example,  $w_2^*$  is the optimal constant prediction that minimizes  $R_2(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^2$

a) In each of the parts below, determine the value of the quantity provided. By “the data”, we are referring to  $y_1, y_2, \dots, y_n$ . The answer choices are as follows; **select one item in each row**.

A: The standard deviation of the data

B: The variance of the data

C: The mean of the data

D: The median of the data

E: The midrange of the data,  $\frac{y_{\min} + y_{\max}}{2}$

F: The mode of the data

G: None of these

|                    | A                     | B                     | C                     | D                     | E                     | F                     | G                     |
|--------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| (i) $w_0^*$        | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| (ii) $w_1^*$       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| (iii) $R_1(w_1^*)$ | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| (iv) $w_2^*$       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| (v) $R_2(w_2^*)$   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

b) Now, suppose we want to find the optimal constant prediction,  $w_U^*$ , using the “Ultra” loss function, defined below:

$$L_U(y_i, w) = y_i(y_i - w)^2$$

To find  $w_U^*$ , we minimize  $R_U(w)$ , the average Ultra loss. How does  $w_U^*$  compare to the mean of the data,  $M$ ?

☐  $w_U^* > M$     ☐  $w_U^* \geq M$     ☐  $w_U^* = M$     ☐  $w_U^* \leq M$     ☐  $w_U^* < M$

- c) Finally, to find the optimal constant prediction, we will instead minimize **regularized** average Ultra loss,  $R_\lambda(w)$ , defined below:

$$R_\lambda(w) = \left( \frac{1}{n} \sum_{i=1}^n y_i (y_i - w)^2 \right) + \lambda w^2$$

Here, assume  $\lambda > 0$  is some positive constant. (We will cover regularization in more detail later in the term.)

Find  $w^*$ , the constant prediction that minimizes  $R_\lambda(w)$ . Give your answer as an expression in terms of the  $y_i$ 's,  $n$ , and/or  $\lambda$ .